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COLDADC_P2(Final) Datasheet

COLDADC is a 16-channel low-noise ADC ASIC intended to read out LArASIC preamps in the DUNE Liquid Argon Far Detectors. The ADC accepts single-ended or differential inputs and outputs a serial data stream to COLDDATA, the DUNE digital data aggregator/serializer chip. The intended sample rate is approximately 2 MSPS. The ASIC produces 16-bit output which is intended to be truncated to 12 or 14 bits.

Revision Number	Key Changes	Drafted by	Revision Date
1.0	Initial draft (Modified from the COLDADC_P1 datasheet)	DCC	7/31/20
1.1	Corrected sections on pipeline stage gain and on use of VCMI.	DCC	8/3/20
1.2	Corrected default values for registers 29 and 30. Included input configuration in description of default setup.	DCC	8/3/20
1.3	Corrected default state of ring oscillator (it is OFF by default).	DCC	8/5/20
1.4	Added figure showing pad numbering.	DCC	9/1/20
1.5	Added note about gap between pads 152 & 153 to figure showing pad numbering.	DCC	9/2/20
1.6	Added conditions for manual calibration. Removed cal_stages[2:0] from register list since it doesn't work.	CRG & DCC	3/22/21
1.7	Corrected figure caption for block diagram.	DCC	5/18/21
1.8	Eliminated redundant (and incomplete) paragraph on manual calibration.	DCC	7/9/21
1.9	Added a note saying that the default value of config_start_number is wrong (default = 0x10, should be 0x0C).	DCC	7/9/21

1.10	Added a note about unbonded pads 170, 172, 174, & 176	DCC	10/26/21
1.11	Changed text to clarify that any value other than 1 in registers 2&3 will power down a SHA	DCC	11/17/21
1.12	Added clarification that VSSED = VSSA2P5	DCC	5/1/22

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Introduction

Each DUNE Front End Motherboard (FEMB) contains eight LArASIC front end ASICs, eight COLDADC ASICs, and two COLDATA ASICs. Each COLDADC ASIC receives input from one LArASIC front end ASIC. The input can be either single-ended or differential. All voltage inputs are sampled simultaneously. Two groups of 8 channels are multiplexed and input to two 15-stage calibrated pipelined ADCs. The ADC outputs are multiplexed onto 8 LVDS channels and sent to COLDATA for further aggregation and transmission via copper links to warm electronics located outside the cryostat. COLDATA is designed to operate at approximately 2 MSPS using a 64 MHz master clock. In DUNE, the master clock will operate at 62.5 MHz and the sample period will be 512 nsec. Rather than 500 nsec.

COLDADC is highly programable and includes two slow control interfaces (UART and “I2C”).

Block Diagram

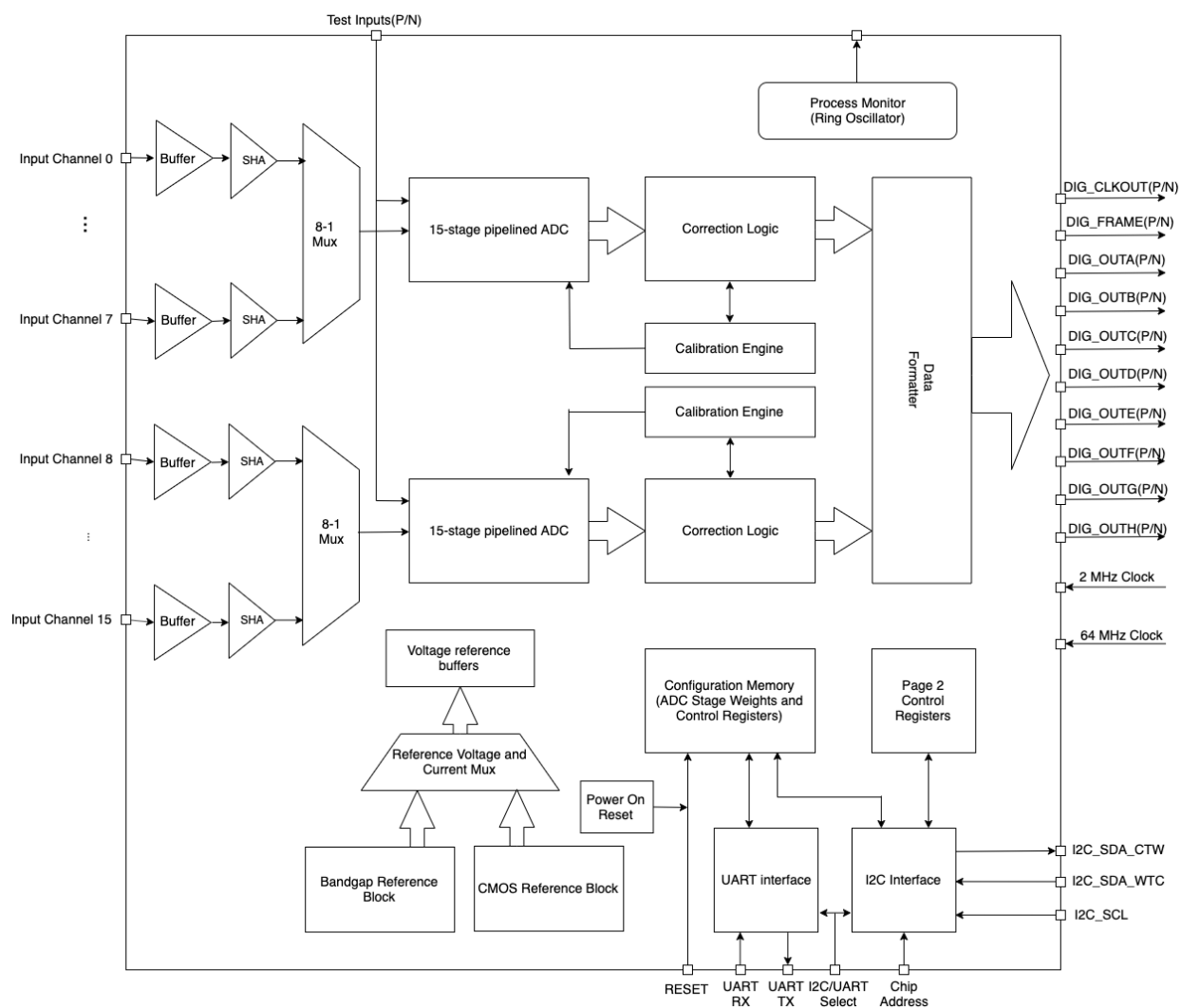


Figure 1: COLDADC_P2 Block Diagram

Power Domains

To minimize coupling between digital and analog circuitry, most of the COLDADC_P2 sub circuits are laid out in deep n-wells. There are four power domains, as shown in Table 1. All of the analog circuitry is powered by VDDA2P5 and resides in a single large deep n-well, connected to VSSA2P5. The digital circuitry that directly controls analog function is powered by VDDD2P5. It also resides in a single deep n-well, connected to VSSD2P5. Most of the digital circuitry is powered by VDDD1P2. It also resides in a single deep n-well, connected to VSSD1P2. The LVDS drivers and CMOS drivers and receivers are powered by VDDIO and reside in separate deep n-wells, each connected to VSSDIO. The pads, ESD protection diodes, and clamps are not in deep n-wells. The protection diodes and clamps are powered by VDDIO and connected directly to the chip substrate (VSSD = VSSA2P5). The LQFP package has a

back contact which is connected to VSSED and should be soldered to the printed circuit board ground.

Power Rail	Value	Units	Current Draw	Note
Analog Power = VDDA2P5	2.25 (+/- 5%)	V	TBD	VDDA2P5 is reduced by 250 mV relative to nominal for thick-oxide devices in to reduce voltage stress (hot electron effects) and increase reliability.
Digital ADC Power = VDDD2P5	2.25 (+/- 5%)	V	TBD	VDDD2P5 is used to power the digital circuits and switches internal to the ADC. VDDD2P5 is reduced by 250 mV relative to nominal to increase reliability.
Digital Logic Power = VDDD1P2	1.1 (+/- 5%) Max voltage is nominal +10%	V	TBD	VDDD1P2 is used to power the synthesized logic core. VDDD1P2 is reduced by 100 mV relative to nominal for thick-oxide devices to reduce voltage stress (hot electron effects) and increase reliability.
ESD Ring / CMOS I/O Power = VDDIO	2.25	V	TBD	VDDES and VDDIO reduced by 250 mV relative to nominal for thick-oxide devices in to reduce voltage stress (hot electron effects) and increase reliability.

Table 1: Power Domains

Functional Description

Input Buffers

As shown in Figure 2, COLDAC has four possible ways to interface with LArASIC. Single-ended inputs can be buffered by single-ended to differential converters (SDCs) or applied directly to the sample-and-hold amplifiers. Similarly, differential inputs can be buffered by differential buffers (DBs) or applied directly to the sample-and-hold amplifiers.

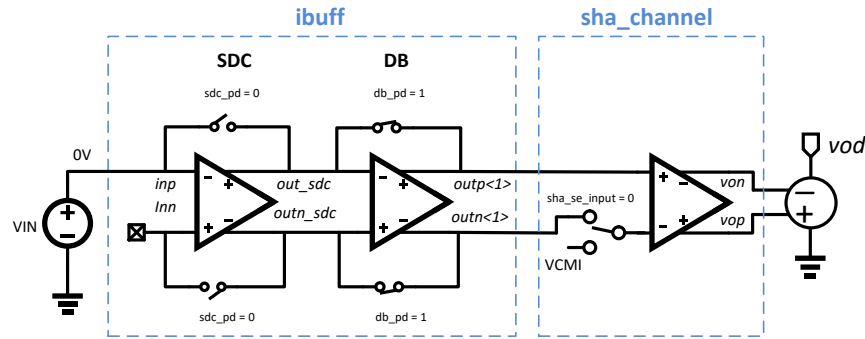


Figure 2: Input Buffer Configuration Options

Sample and Hold Amplifiers and Multiplexer

The Sample and Hold Amplifiers (SHAs) are organized in two groups of eight. They sample the input voltages simultaneously at a time controlled by the “2 MHz” clock from COLDATA. The internal 2 MHz clock is resynchronized to the “64” MHz clock and a synchronous “16 MHz” clock is generated and used to clock an 8-to-1 analog multiplexer that presents 8 samples in turn to each of the two ADC pipelines. If the phase of the 2 MHz clock is changed, output data will be garbled for one or two sample periods.

Pipelined ADC and Correction Logic

A conceptual diagram of an ADC pipeline is shown in Figure 3. Each of the 15 pipeline stages contains a low-resolution 1.5-bit analog-to-digital subconverter containing two comparators, a 1.5-bit digital-to-analog subconverter that produces a voltage based on the two comparator outputs, an analog subtractor, a sample-and-hold amplifier, and an interstage 2x amplifier. Each pipeline stage has the same transfer function (shown in Figure 4). The comparator outputs determine both the digital output of the stage (W_0 , 0, or W_2) and the voltage that is subtracted from the analog value input to the stage before the difference is amplified and input to the next stage. The 16-bit output of the pipeline is the sum of the digital outputs from each of the pipeline stages. The “stage weights” (W_0 and W_2) are 16-bit two’s complement (signed) integers.

The output of each analog pipeline stage is 0, 1, or 2 depending on the comparator outputs. The correction logic sums the corresponding “weights” and produces a 16-bit ADC output. It is possible to test the correction logic by setting the control bit *test_correction_logic* in register 32 and loading 30-bit patterns into registers 33-40.

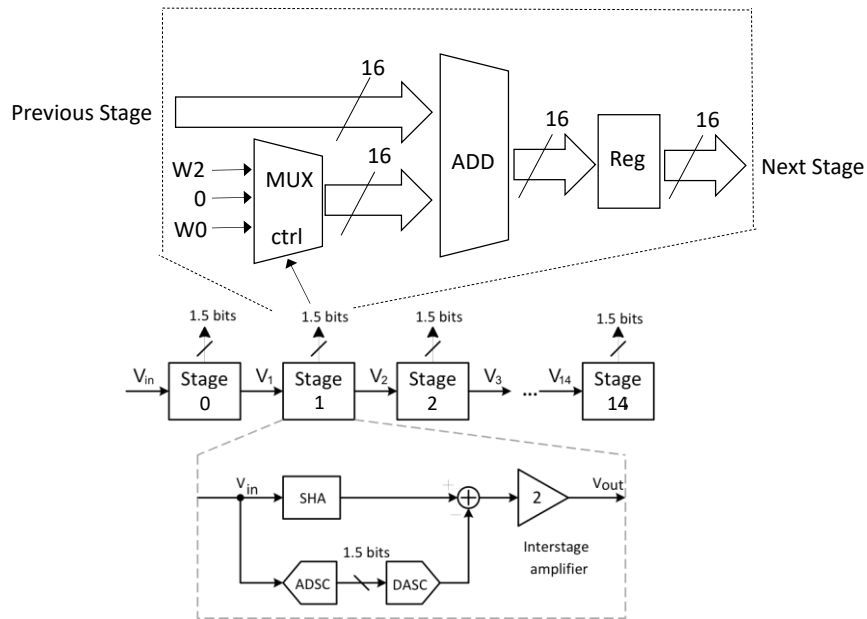


Figure 3: Pipelined ADC. Each pipeline stage contains a Sample and Hold Amplifier (SHA), an Analog to Digital SubConverter (ADSC) consisting of two comparators, a Digital to Analog SubConverter (DASC) that produces a voltage based on the ADSC output, an analog subtracter, and a gain stage.

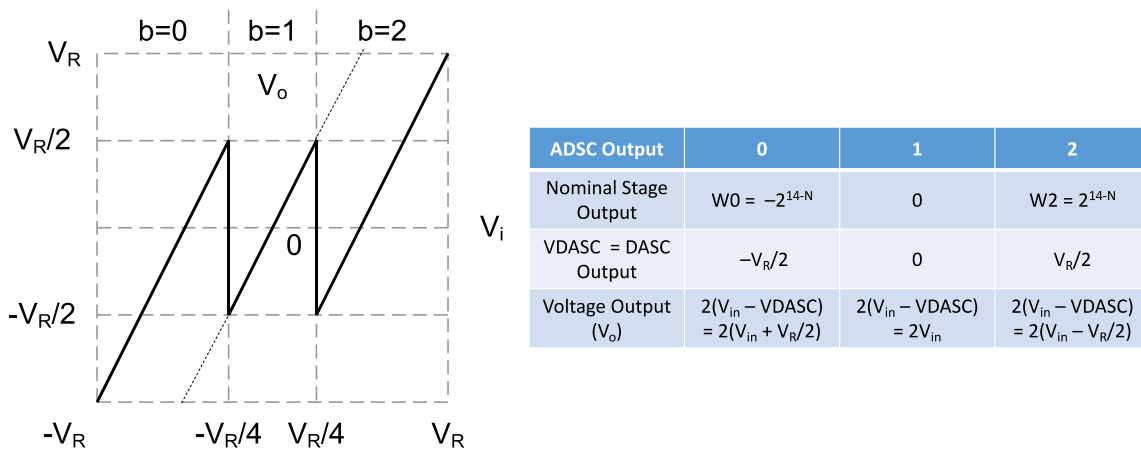


Figure 4: ADC pipeline stage transfer function.

Calibration Engine

The Calibration Engine allows the correction of nonlinearities caused by imperfections in the voltages that are passed from one stage to the next. These imperfections arise from errors in the analog voltage subtracted in each stage and in the gain and offset of the interstage amplifiers.

COLDADC is calibrated using a bootstrap procedure that relies on the fact that the required precision is easily satisfied by the last stages of the pipeline. The seven most significant stages are calibrated by the automated procedure. Starting with the least significant stage to be calibrated, the input to the stage being calibrated is set to the threshold levels of $\pm V_R/4$ and the normal comparator outputs are overridden and forced first to 1 and then to 0. The lower stages of the ADC digitize the analog value output from the stage being calibrated and the difference between the ADC output when the comparator is forced to 1 and when it is forced to 0 is calculated. These two differences (W0 expressed as a negative number and W2 expressed as a positive number) are stored and used as two of the three possible outputs of the stage being calibrated (the other possible output being 0). This procedure is then repeated for the next most significant pipeline stage until stage 0 has been calibrated.

In order to eliminate sensitivity to noise in the individual ADC conversions during calibration, each value is calculated a number of times and an average value used. The number of samples used is set by *meas_cycles*[3:0] in register 32.

The calibration procedure for ADC0 is started by setting bit 0 of *calibrate*[1:0] in register 31. Setting bit 1 of *calibrate*[1:0] starts the automatic calibration of ADC1. Normal ADC operation resumes as soon as the calibration procedure is completed.

The default weights (shown in the table on the right of Figure 4) are computed for an interstage gain of exactly two. If the actual interstage gain is greater than two, the digital value computed for input voltages close to the extreme values may require 17 bits. An overflow monitor is included in the correction logic that detects this condition and sets the output to the minimum or maximum ADC value as appropriate. The overflow monitor can be disabled by setting *enable_overflow_mon* in register 41 to 0.

It is possible to verify the automatic calibration procedure by using registers 43 – 46 to force the configuration of the pipeline. In this case, the differences and averages described above must be calculated off chip and the W0 and W2 values updated before the calibration of the next most significant pipeline stage is begun. When the calibration procedure is executed in this fashion, the overflow monitor must be disabled by setting *enable_overflow_mon* in register 41 to 0.

Data Formatter

The Data Formatter receives data from the two ADC pipelines and formats it for output. Data is output using 8 LVDS pairs. Four pairs are used for data from ADC0 and four pairs are used for data from ADC1. Each pair carries data from one nibble of a 16-bit ADC value as shown in

Table 2. Data is output most significant bit first using a “64” MHz clock to match the rate that analog values are presented to the input of the ADC pipelines for digitization. Output data changes value on the rising edge of DIG_CLKOUT and is latched at COLDATA on the falling edge of DIG_CLKOUT. The start of a block of data corresponding to a common sample time is indicated by DIG_FRAME. The channel order is 0 – 7 on outputs A – D and 8 – 15 on outputs E – H. If DIG_FRAME is misplaced, it can be moved in time with respect to the output data using *config_start_number[4:0]* in register 49 (or page 2, register 1 if *I2C_UART_SEL* = 0). If necessary, the order in which input channels are presented to the ADC pipelines can also be adjusted using *config_SHA0_Pointer[2:0]* in register 49 (or page 2, register 4 if *I2C_UART_SEL* = 0).

Serial Data Output	ADC Output Nibble
DIG_OUT_A	ADC0[3:0]
DIG_OUT_B	ADC0[7:4]
DIG_OUT_C	ADC0[11:8]
DIG_OUT_D	ADC0[15:12]
DIG_OUT_E	ADC1[3:0]
DIG_OUT_F	ADC1[7:4]
DIG_OUT_G	ADC1[11:8]
DIG_OUT_H	ADC1[15:12]

Table 2: Relationship between output data ports and ADC outputs.

Reference Blocks

COLDADC contains redundant circuit blocks for reference voltage and bias current generation. The control variables *bgr_select* in register 19 and *iref_sel[1:0]* in register 28 determine whether the bandgap reference block or the CMOS reference block is used (note that both control variables must be set). The CMOS reference block is designed for use with an internal reference resistor but an external resistor can also be used. It is also possible to override both reference blocks and use a simple PMOS current mirror connected directly to an external resistor. This operating mode will have significantly poorer power supply rejection and is included as part of the CMOS reference block as backup only.

Reference Voltages

Four reference voltages are used in COLDADC. VCM1 is used as the reference voltage (either in the input buffers or in the SHAs if the input buffers are bypassed) when single ended inputs are being used. It is also used by the SHAs to precharge the output stage of the SHAs while the inputs are being sampled. VCM0 defines the “zero” value for the pipeline ADCs. It is the common mode voltage for the output of the differential input buffers and of the SHAs. VREFN and VREFP are the negative and positive rails of the pipeline ADCs. The nominal values for the various reference voltages are summarized in Table 3.

Reference Name	Value	Units	Description
VREFP	1.95	V	ADC positive reference voltage
VREFN	0.45	V	ADC negative reference voltage
VCMI	0.9	V	Common-mode input voltage. Used in input buffer and SHA.
VCMO	1.2	V	ADC Core Common-mode voltage. Also used in SHA.

Table 3: Nominal reference voltages. Note that V_R in Figure 4 is $V_R = (VREFP - VREFN)/2 + VCMO$.

The BJT-based reference requires filter capacitors at the output of the reference (before the buffers). The CMOS-based reference does not require filter capacitors. If operation of the ADC with external references is desired, the two reference generators can be powered down and references defined off chip can be introduced into the circuit through dedicated pads prior to the ADC reference buffers. The output of the MUX goes to buffers that drive the ADC reference voltage inputs.

Reference Currents

The input buffers require two current sources (both nominally $200\ \mu\text{A}$). Control bits in register 0 (*ibuff_ctrl*[3:0]) select which reference block sources the currents and which set of input buffers receives the currents. The current magnitudes are controlled separately depending on the current source.

Each bank of 8 SHAs and each pipeline ADC requires a $50\ \mu\text{A}$ current sink. The voltage reference buffer block is implemented in two sections, one for channels 0 – 7 and one for channels 8 – 15. Each block requires a $250\ \mu\text{A}$ current sink. There are no control bits associated with the current sinks.

The SHA bias circuit uses its $50\ \mu\text{A}$ current sink to produce 8 independent current sources. The magnitude of these current sources are controlled using control bits in registers 4 – 7. The currents for SHA0 and SHA7 share a control variable (*sha0_bias*[2:0]). Similarly the magnitude of the bias current for SHA1 and SHA8 is controlled by *sha7_bias*[2:0], and so on. By default all of the SHA bias currents are equal. Since SHA7 can be much slower than SHA0, power can be saved by reducing the current to some of the SHAs.

The ADC bias circuit uses a $50\ \mu\text{A}$ current sink and produces a single bias current for the 14 interstage amplifiers (OP amps) in the pipeline (the least significant pipeline stage does not include an OP amp). The magnitude of the OP amp bias current for both ADC pipelines controlled by *adc_bias*[2:0] in register 8.

Finally, the voltage reference buffer circuits use a $250\ \mu\text{A}$ current sink and produce current sources for the reference voltage buffer amplifiers. These are controlled by a single control variable, *ref_bias*[2:0], in register 23.

Bandgap Reference

The bandgap reference block uses two diode connected vertical pnp transistors with different current densities to create a reference voltage (V_{BGR}) that depends only weakly on the bias voltage and on temperature. This voltage is used to generate a unit current that is used by digital to analog converters to create programmable reference voltages and currents. The four reference voltages are controlled by variables in registers 10 – 14 and may be set in 8 mV steps.

In addition to the current sources and sinks described above, the bandgap reference block creates two nominally 200 μA current sources that are used by amplifiers in the digital to analog converters that produce the reference voltages. The magnitude of these currents may be adjusted using control variables in registers 17 and 18.

The magnitude of the currents provided to the input buffers may be adjusted using control variables in registers 15 and 16.

CMOS Reference

The CMOS reference block depends on a master bias current whose accuracy is expected to be poor relative to the bandgap voltage used by the bandgap reference block. For this reason, trimming bits are included to be used in conjunction with a current monitor output to allow the user to set the master bias current to 50 μA . The trimming bits (*vt_ref_trim_ctrl*[2:0] in register 28) allow the CMOS reference bias current to be adjusted in steps of 5 μA .

The voltages produced by the CMOS reference block are directly proportional to the bias voltage (V_{DDA2P5}) and are controlled by variables in registers 24 – 27. The four reference voltages may be set in 8.8 mV steps. The currents provided to the input buffers are controlled by variables in registers 29 and 30.

Control Interface

All COLDADC_P2 control I/O uses CMOS signals and is in the VDDIO domain. The logic levels are 1 = 2.25V and 0 = 0V.

Most of the COLDADC_P2 configuration registers can be read or written using either a custom UART or an I2C-like interface. To use the UART, *I2C_UART_SEL* (pad a, pin b) must be connected to 2.25V. To use the I2C interface, *I2C_UART_SEL* must be connected to 0V. The UART interface can be used to read or write all registers in the configuration memory. It cannot access the “page 2” back end control registers. For this reason, if *I2C_UART_SEL* is connected to 2.25V, the page 2 back end control registers are disabled, and back end functions are controlled by variables in registers 48 – 50.

Both control interfaces use a chip address that is set using wire bond pads *I2C_ADD_0* – *I2C_ADD_3* (or package pins *I2C_ADD*<0> – *I2C_ADD*<3>). As above, 2.25V = 1 and 0V = 0.

I2C

The “I2C-like” interface implemented in COLDADC_P2 is similar to classical, single-master I2C with no clock extension. The major difference is motivated by the fact that the DUNE I2C communications must travel over long cables between the WIB and the FEMBs. Consequently, for these signals, canonical I2C signaling is replaced by LVDS. However, LVDS is not amenable to bidirectional communication, so canonical I2C’s bidirectional SDA (Serial Data) line is replaced by two data lines, one from Warm to Cold (SDA_w2c) and one from Cold to Warm (SDA_c2w). In order to reduce power dissipation on the FEMBs, CMOS signaling rather than LVDS is used for I2C communication on the FEMB.

Other differences between DUNE I2C and canonical I2C are that DUNE I2C does not include block transfers, and it uses a 3-word protocol rather than a 2-word protocol. DUNE I2C commands are composed of three bytes of information input serially to the chip on I2C_SDA_w2c. Nine clock pulses are issued for each byte. The first 8 pulses are used to shift data either into or out of the ASIC. The I2C interface acknowledges the transmission by raising the I2C_SDA_c2w line on the ninth clock. The rising edges of data on both SDA lines are roughly in time with the rising clock edges. Data should be latched at its destination using the falling edge of the I2C_SCL clock. Bytes are shifted in most significant bit first. The first byte includes the chip address (4 bits), the register page address (3 bits), and a single bit that is set to 0 for a write command and 1 for a read command. The second byte is the 8-bit register address. For a read command, the contents of the addressed register are read out on the I2C_SDA_c2w line during the third byte. For a read command, the contents of the I2C_SDA_w2c line are unimportant during the third byte. For a write command, the third byte is the data to be written and it is presented to the chip on the I2C_SDA_w2c line. The Soft Reset command has the format of a Write to chip address 0, register page 0, register address 6: ([00000000][00000110]). The contents of the third byte are unimportant. The Soft Reset restores all page 2 control registers to their default values. It does not affect the registers in the configuration memory (page 1 registers).

The 64 MHz clock is required for the I2C interface to work; internal state machines run on the 64 MHz clock. The I2C_SCL frequency must be between ~0.5 MHz and 2 MHz. If the chip address does not match the address set by *I2C_ADD_0* – *I2C_ADD_3*, the response on I2C_SDA_c2w will be 0x5A. The figures below illustrate I2C communications.

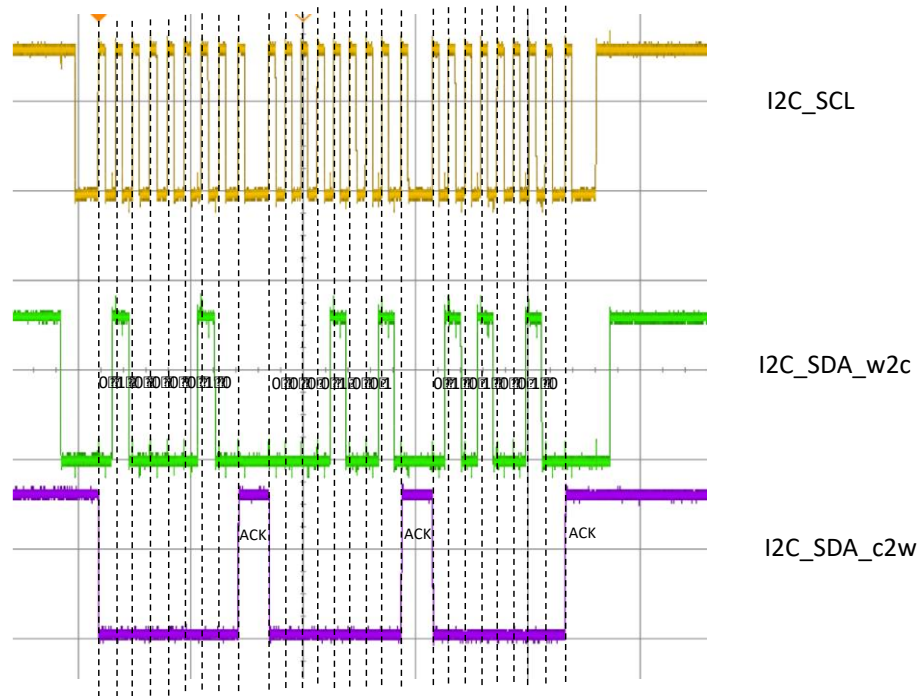


Figure 5: I2C write format; this example is write(chip 4, page 1); register 9; 0x52.

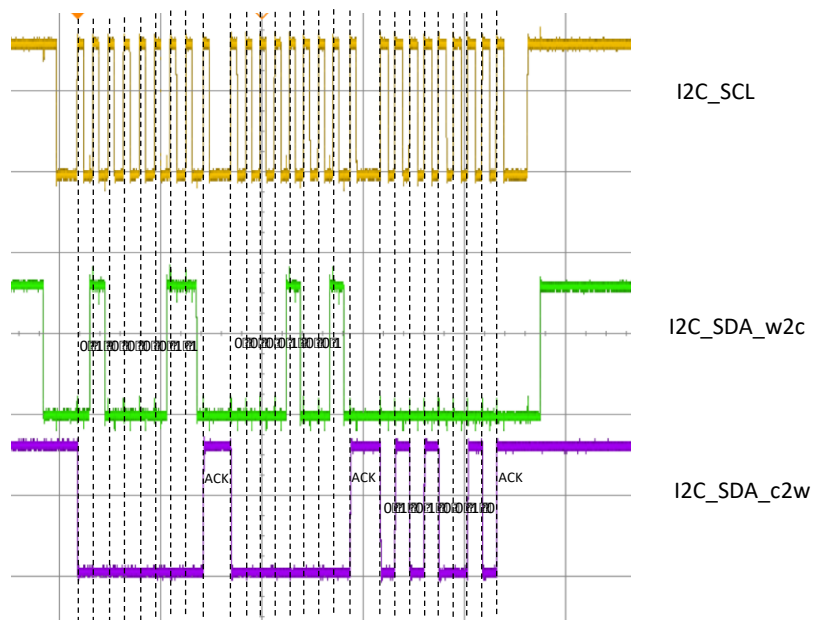


Figure 6: I2C read format; this example is read(chip 4, page 1); register 9. The response is 0x52.

In order to simplify the logic used to relay I2C information through COLDATA data concentrator chips on the DUNE FEMBs, a chip addressing convention (shown below) has been established. The I2C interface in COLDADC_P2 will not read or write page 2 registers unless the COLDADC_P2 chip address is 01xx or 10xx. However, the page 1 registers can be written and read using the I2C interface for all possible chip addresses.

Chip Address	Assignment
0000 – 0001	Unused
0010	“Bottom” COLDATA
0011	“Top” COLDATA
01xx (xx = 00,01,10,11)	COLDADCs attached to COLDATA_BOT
10xx (xx=00,01,10,11)	COLDADCs attached to COLDATA_TOP
11xx (xx=00,01,10,11)	Unused

Table 4: COLDATA I2C chip address convention.

UART

COLDADC_P2 includes a simple custom Universal Asynchronous Receiver/Transmitter (UART). The UART uses a single CMOS pin for input (*UART_RX*) and a single CMOS pin for output (*UART_TX*). The UART operates with a 22-bit word length, an 8-bit address space and an 8-bit data payload. The UART bits are summarized in Table 5.

Bit Range	Contents	Comment
[0]	WRB	1 to write, 0 to read
[4:1]	Chip ID	Shared with I2C bus
[12:5]	Data	8-bit data payload
[20:13]	Address	8-bit address
[21]	Parity	0 if number of 1s in bits [20:0] is odd (odd parity). Ignored for write operations.

Table 5: UART bit definitions.

When a UART transaction is requested externally (by shifting bits into the *UART_RX* pin) the first bit in the UART word is the WRB indicator. When WRB is high, a write is being requested and the payload byte will be written to the memory location contained in the address byte. When WRB is low, a read is being requested and the payload byte located in the memory location contained in the address byte will be sent out the *UART_TX* pin.

The logical interface of the PHY is shown in Figure 7. When idle, the *UART_TX* output is high. The first bit of a UART transaction is a start bit (always a low bit), and the last is a stop bit (always a high bit). The data bits are sent LSB-first.



Figure 7: UART output word including start and stop bits.

The UART is internally oversampled by a factor of 16 to guard against triggering of a transaction by noise on the *UART_RX* pin. The timing diagram for the UART is shown in Figure 8. The UART is clocked with the ADC input clock (which is generated internal to the Cold ADC). Therefore, UART data should be sent to Cold ADC at approximately 1 MHz (assuming a 64 MHz master chip clock). The output data on *UART_TX* is clocked using the internal 16 MHz clock. This means the transmit frequency will be slightly less than 1 MHz if the 64 MHz clock is actually 62.5 MHz as planned.

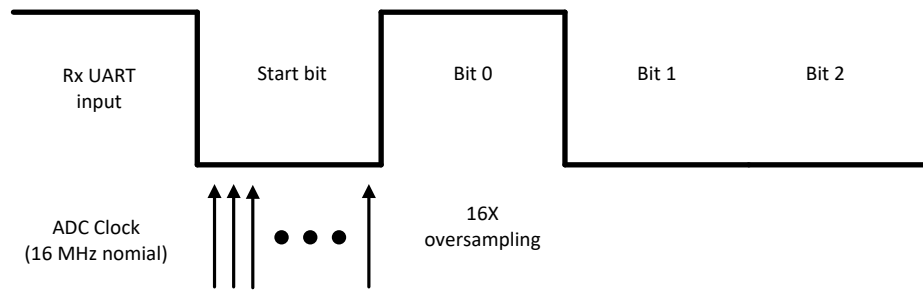


Figure 8: UART timing diagram.

Power On Reset

The power on reset circuit is intended to initialize all control variables to their default values. The *POR_NAND* input will disable the power on reset circuit if tied to 0V. An analog pad (*POR_SF*) is included for debugging purposes. The power on reset circuit will not operate properly if the rise time of *VDDA2P5* is longer than ~1 msec.

Process Monitor

COLDADC_P2 contains a process monitor that is intended to help characterize the process variations associated with different wafers and different fabrication lots. The process monitor is a ring oscillator comprising a chain of 40 inverters and one NAND gate followed by a divide-by-32 stage (to reduce the frequency that must be driven off-chip). The ring oscillator is powered by *VDDD1P2* and enabled by *enable_ringosc* in register 42. By default, the ring oscillator is off. Table 6 shows the expected oscillation frequency as a function of process corner and temperature. The expected current consumption is shown in Table 7.

Process Corner	-200 C	27 C
Slow-Slow	18.4 MHz	16.1 MHz
Typical-Typical	24.2 MHz	18.9 MHz
Fast-Fast	34.9 MHz	22.5 MHz

Table 6: Oscillation Frequency

	-200 C	27 C
Average	397 μ A	325 μ A
Peak	982 μ A	650 μ A

Table 7: Average and peak current drawn by ring oscillator.

The COLDADC_P2 die size is 6860 microns by 7610 microns. There are 178 wire bond pads, 47 on the left and right sides (inputs from LArASIC and ADC outputs), and 42 on the top and bottom. The low profile quad flat package has 128 pins, 32 on each side, and a back-side contact. The LQFP package is 14 mm by 14 mm and 1.6 mm high. Figure 9 shows the wire bonding pad numbering. Table 8 lists the package pin numbers as well as the pad numbers.



Pad #	Pad Name	Pin #	Pin Name	Comment
1	INP[15]	1	INP<15>	
2	INN[15]	2	INN<15>	
3	VSSA2P5		VSSESD	Down bond
4	INP[14]	3	INP<14>	
5	INN[14]	4	INN<14>	
6	VSSA2P5		VSSESD	Down bond
7	INP[13]	5	INP<13>	
8	INN[13]	6	INN<13>	
9	VSSA2P5		VSSESD	Down bond
10	INP[12]	7	INP<12>	
11	INN[12]	8	INN<12>	
12	VSSA2P5		VSSESD	Down bond
13	INP[11]	9	INP<11>	
14	INN[11]	10	INN<11>	
15	VSSA2P5		VSSESD	Down bond
16	INP[10]	11	INP<10>	
17	INN[10]	12	INN<10>	
18	VSSA2P5		VSSESD	Down bond
19	INP[9]	13	INP<9>	
20	INN[9]	14	INN<9>	
21	VSSA2P5		VDDESD	Down bond
22	INP[8]	15	INP<8>	
23	INN[7]	16	INN<8>	
24	VSSA2P5		VSSESD	Down bond
25	INP[7]	17	INP<7>	
26	INN[7]	18	INN<7>	
27	VSSA2P5		VSSESD	Down bond
28	INP[6]	19	INP<6>	
29	INN[6]	20	INN<6>	
30	VSSA2P5		VSSESD	Down bond
31	INP[5]	21	INP<5>	
32	INN[5]	22	INN<5>	
33	VSSA2P5		VSSESD	Down bond
34	INP[4]	23	INP<4>	
35	INN[4]	24	INN<4>	
36	VSSA2P5		VSSESD	Down bond
37	INP[3]	25	INP<3>	
38	INN[3]	26	INN<3>	
39	VSSA2P5		VSSESD	Down bond

40	INP[2]	27	INP<2>	
41	INN[2]	28	INN<2>	
42	VSSA2P5		VSSED	Down bond
43	INP[1]	29	INP<1>	
44	INN[1]	30	INN<1>	
45	VSSA2P5		VSSED	Down bond
46	INP[0]	31	INP<0>	
47	INN[0]	32	INN<0>	
48	VDDA2P5	33	VDDA2P5	
49	VSSA2P5	34	VSSA2P5	
50	VDDA2P5	35	VDDA2P5	
51	VDDA2P5		VDDA2P5	Double bond to #35
52	VSSA2P5	36	VSSA2P5	
53	VSSA2P5		VSSA2P5	Double bond to #36
54	VDDA2P5	37	VDDA2P5	
55	VDDA2P5		VDDA2P5	Double bond to #37
56	VSSA2P5	38	VSSA2P5	
57	VSSA2P5		VSSA2P5	Double bond to #38
58	VDDD2P5	39	VDDD2P5	
59	VSSD2P5	40	VSSD2P5	
60	VDDD2P5	41	VDDD2P5	
61	VDDD2P5		VDDD2P5	Double bond to #41
62	VSSD2P5	42	VSSD2P5	
63	VSSD2P5		VSSD2P5	Double bond to #42
64	VSSA2P5	43	VSSA2P5	
65	VDDA2P5	44	VDDA2P5	
66	VDDA2P5		VDDA2P5	Double bond to #44
67	VSSA2P5	45	VSSA2P5	
68	VSSA2P5		VSSA2P5	Double bond to #45
69	AUX_ISINK	46	AUX_ISINK	Also called aux 2
70	AUX_VOLTAGE	47	AUX_VOLTAGE	Also called aux 1
71	AUX_ISOURCE	48	AUX_ISOURCE	Also called aux 3
72	VREF_EXT	49	VREF_EXT	
73	VREF_DECOUPLE	50	VREF_DECOUPLE	
74	RBIAS_CMOS	51	RBIAS_CMOS	
75	VREFP	52	VREFP	
76	VREFN	53	VREFN	
77	VCMI	54	VCMI	
78	VCMO	55	VCMO	
79	VSSA2P5	56	VSSA2P5	
80	ADC_TEST_IN_P	57	ADC_TEST_IN_P	

81	ADC_TEST_IN_N	58	ADC_TEST_IN_N	
82	VSSA2P5	59	VSSA2P5	
83	VOLTAGE_MONITOR	60	VOLTAGE_MONITOR	
84	CURRENT_MONITOR	61	CURRENT_MONITOR	
85	<i>unconnected</i>		<i>no name</i>	
86	VSSA2P5	62	VSSA2P5	
87	VSSA2P5		VSSA2P5	Double bond to #62
88	VDDD1P2	63	VDDD1P2	
89	VSSD1P2	64	VSSD1P2	
90	VSSIO	65	VSSIO	
91	VDDIO	66	VDDIO	
92	DIG_OUTH_N	67	DIG_OUTH_N	
93	DIG_OUTH_P	68	DIG_OUTH_P	
94	VSSA2P5		VSSA2P5	Down bond
95	DIG_OUTG_N	69	DIG_OUTG_N	
96	DIG_OUTG_P	70	DIG_OUTG_P	
97	VSSA2P5		VSSA2P5	Down bond
98	DIG_OUTF_N	71	DIG_OUTF_N	
99	DIG_OUTF_P	72	DIG_OUTF_P	
100	VSSA2P5		VSSA2P5	Down bond
101	DIG_OUTE_N	73	DIG_OUTE_N	
102	DIG_OUTE_P	74	DIG_OUTE_P	
103	VSSA2P5		VSSA2P5	Down bond
104	DIG_OUTD_N	75	DIG_OUTD_N	
105	DIG_OUTD_P	76	DIG_OUTD_P	
106	VSSA2P5		VSSA2P5	Down bond
107	DIG_OUTC_N	77	DIG_OUTC_N	
108	DIG_OUTC_P	78	DIG_OUTC_P	
109	VSSA2P5		VSSA2P5	Down bond
110	DIG_OUTB_N	79	DIG_OUTB_N	
111	DIG_OUTB_P	80	DIG_OUTB_P	
112	VSSA2P5		VSSA2P5	Down bond
113	DIG_OUTA_N	81	DIG_OUTA_N	
114	DIG_OUTA_P	82	DIG_OUTA_P	
115	VSSA2P5		VSSA2P5	Down bond
116	DIG_FRAME_N	83	DIG_FRAME_N	
117	DIG_FRAME_P	84	DIG_FRAME_P	
118	VSSA2P5		VSSA2P5	Down bond
119	DIG_CLKOUT_N	85	DIG_CLKOUT_N	
120	DIG_CLKOUT_P	86	DIG_CLKOUT_P	
121	VSSIO	87	VSSIO	

122	VDDIO	88	VDDIO	
123	CLK_2MHZ_N	89	CLK_2MHZ_N	
124	CLK_2MHZ_P	90	CLK_2MHZ_P	
125	VSSA2P5		VSSSED	Down bond
126	CLK_64MHZ_N	91	CLK_64MHZ_N	
127	CLK_64MHZ_P	92	CLK_64MHZ_P	
128	VSSA2P5		VSSSED	Down bond
129	VSSIO	93	VSSIO	
130	VSSIO	94	VSSIO	
131	VSSIO		VSSIO	Double bond to #94
132	VDDIO		VDDIO	Double bond to #95
133	VDDIO	95	VDDIO	
134	VDDIO	96	VDDIO	
135	DIGITAL_MUX_OUT_N			not connected to pin
136	DIGITAL_MUX_OUT_P			not connected to pin
137	POR_NAND	97	POR_NAND	
138	POR_SF			not connected to pin
139	VSSA2P5		VSSSED	Down bond
140	CHIP_ACTIVE	98	CHIP_ACTIVE	
141	RO_OUT	99	RO_OUT	Ring Oscillator Out
142	I2C_ADD_3	100	I2C_ADD<3>	
143	I2C_ADD_2	101	I2C_ADD<2>	
144	I2C_ADD_1	102	I2C_ADD<1>	
145	I2C_ADD_0	103	I2C_ADD<0>	
146	I2C_SCL	104	I2C_SCL	
147	I2C_SDA_W2C	105	I2C_SDA_W2C	
148	I2C_SDA_C2W	106	I2C_SDA_C2W	
149	I2C_UART_SEL	107	I2C_UART_SEL	Pull High to use UART
150	UART_RX	108	UART_RX	
151	UART_TX	109	UART_TX	
152	MASTER_RESET	110	MASTER_RESET	Resets Config Memory
153	VSSA2P5	111	VSSSED	
154	VSSD1P2	112	VSSD1P2	
155	VDDD1P2	113	VDDD1P2	
156	VSSD1P2	114	VSSD1P2	
157	VSSD1P2		VSSD1P2	Double bond to #114
158	VDDD1P2	115	VDDD1P2	
159	VDDD1P2		VDDD1P2	Double bond to #115
160	VSSD1P2	116	VSSD1P2	
161	VDDD1P2	117	VDDD1P2	
162	VSSA2P5	118	VSSSED	

163	VSSD2P5	119	VSSD2P5	
164	VSSD2P5		VSSD2P5	Double bond to #119
165	VDDD2P5	120	VDDD2P5	
166	VDDD2P5		VDDD2P5	Double bond to #120
167	VSSD2P5	121	VSSD2P5	
168	VDDD2P5	122	VDDD2P5	
169	VSSA2P5	123	VSSA2P5	
170	VSSA2P5		VSSA2P5	Double bond to #123
171	VDDA2P5	124	VDDA2P5	
172	VDDA2P5		VDDA2P5	Double bond to #124
173	VSSA2P5	125	VSSA2P5	
174	VSSA2P5		VSSA2P5	Double bond to #125
175	VDDA2P5	126	VDDA2P5	
176	VDDA2P5		VDDA2P5	Double bond to #126
177	VSSA2P5	127	VSSA2P5	
178	VDDA2P5	128	VDDA2P5	

Table 8: Wirebond Pad and Package Pin List

Note: The first COLDAC_P2 chips packaged (by ASE with labels including “DUNE”) had pads 170, 172, 174, and 176 unbonded. The parts from the first 4 wafers of the engineering run were packaged by Gretek (GTK) and will be labeled simply “COLDAC” (without serial numbers). These parts have pads 170, 172, 174, and 176 bonded as indicated above.

Configuration Memory

The configuration memory holds the pipeline stage weights (W_0 and W_2) and a number of control registers (the “page 1 registers”). The control registers are 8-bit registers while the pipeline stage weights (and pipeline gains and offsets) are 16-bit numbers. The memory is accessed one byte at a time. 16-bit numbers are stored in two consecutive memory locations, with low order 8 bits in an address that is one less than the high order 8 bits. A block diagram of the configuration memory is shown in Figure 10. The configuration memory can be accessed using either the UART or the I2C interface. If the I2C interface is used, the page number needs to be set = 1.

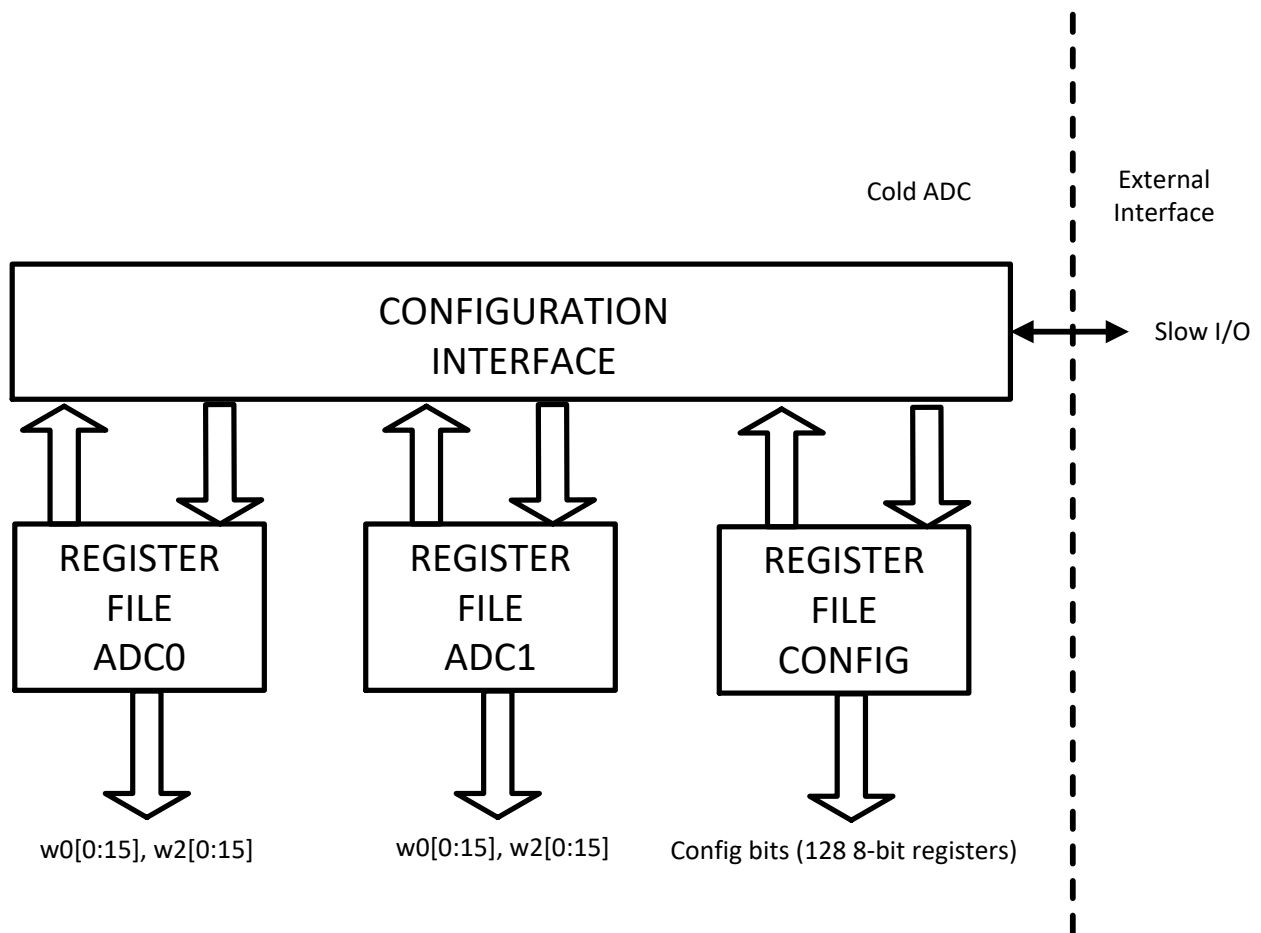


Figure 10: Configuration Memory

The memory map for the configuration memory is shown in Table 9.

Address Range (Hexadecimal)	Contents	Notes
FF-80	Configuration Bits	See section on configuration settings.
7F-7E	ADC1 Offset	
7D-60	ADC1, W2	
5F-5E	ADC1 Gain	
5D-40	ADC1, W0	
3F-3E	ADC0 Offset	
3D-20	ADC0, W2	
1F-1E	ADC0, Gain	
1D-00	ADC0, W0	

Table 9: Configuration Memory Map

With this memory map, the address of a particular element can be assembled using the table below.

Bit	[7]	[6]	[5]	[4:1]	[0]
Contents	config flag 0 = Weights 1 = Control registers	which ADC 0 = ADC0 1 = ADC1	which weight 0 = W0 1 = W1	stage address 0 = Most significant	which byte 0 = least significant byte

Table 10: Configuration Memory Address

The stage weights are 16-bit numbers and hence each occupies two memory locations. The memory address for a stage weight can be constructed by setting bit 7 = 0, bit 6 = 0 for ADC0 or 1 for ADC1, bit 5 = 0 for W0 or 1 for W1, bits [4:1] = pipeline stage number (0000 for the first pipeline stage and 1111 for the last pipeline stage), and bit 0 = 0 for the least significant byte or 1 for the most significant byte. The control registers can be addressed by setting bit 7 = 1 and bits [6:0] = the register number. That is, the address of each control register is the register number + 128.

The default stage weights are given in Table 11.

ADC Stage	W0	W2	ADC Stage	W0	W2
0	0xC000	0x4000	7	0xFF80	0x0080
1	0xE000	0x2000	8	0xFFC0	0x0040
2	0xF000	0x1000	9	0xFFE0	0x0020
3	0xF800	0x0800	10	0xFFFF0	0x0010
4	0xFC00	0x0400	11	0xFFFF8	0x0008
5	0xFE00	0x0200	12	0xFFFFC	0x0004
6	0xFF00	0x0100	13	0xFFFFE	0x0002
			14	0xFFFF	0x0001

Table 11: Default ADC pipeline stage weights.

Page 1 Control Registers

Input Buffer Configuration

Register	Control Name	Description	Bits	Default = 0xA3
0 0x80	ibuff_sdc_pd	0 → Power up single-to-differential converter in the input buffer 1 → Power down and bypass of single-to-differential converter in the Input Buffer.	[0]	0b1
	ibuff_diff_pd	0 → Power up differential buffer 1 → Power down and bypass of differential buffer in the Input buffer Note: When powering down both single ended and differential input buffers, it is necessary to turn all ibuff bias currents down to minimum value.	[1]	0b1
	ibuff_ctrl[3:0]	Control of current multiplexer. BJT or CMOS reference. Directs current to SDC or DB: ibuff_ctrl[0] = 1 directs current from Bandgap Reference to input buffers ibuff_ctrl[1] = 1 directs current from CMOS Reference to input buffers ibuff_ctrl[2] = 1 directs currents to single ended input buffers (SDC) ibuff_ctrl[3] = 1 directs currents to differential input buffers (DB)	[7:4]	0xA

Care should be taken not to power up both the single-to-differential converter and the differential buffers. In the case when both the single-to-differential converter and the differential buffer are bypassed care must be taken to set the configuration bit *sha_se_input* (in register 4) high if the input to Cold ADC is single-ended (and a conversion to differential in the SHA is desired).

Sample and Hold (SHA) Amplifier Configuration

Register	Control Name	Description	Bits	Default = 0x00
1 0x81	freeze_sha[1:0]	0 → normal operation 1 → freezes SHA output multiplexer Bit 0 corresponds to ADC0 and bit 1 corresponds to ADC1	[1:0]	0x0
	freeze_select0[2:0]	Selects which SHA is connected to ADC0 when frozen. 000 = SHA0, 001 = SHA1, etc.	[4:2]	0x0
	freeze_select1[2:0]	Selects which SHA is connected to ADC1 when frozen. 000 = SHA0 (channel 8), 001 = SHA1 (channel 9), etc.	[7:5]	0x0

Register	Control Name	Description	Bits	Default = 0x00
2 0x82	sha_pd_0	Power down SHAs associated with ADC0 0 → normal operation Any other value → power down SHAs	[0]	0x0

Register	Control Name	Description	Bits	Default = 0x00
3 0x83	sha_pd_1	Power down SHAs associated with ADC1 0 → normal operation Any other value → power down SHAs	[0]	0x0

Register	Control Name	Description	Bits	Default = 0x33
4 0x84	sha0_bias	Bias adjustment for SHA0. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	[2:0]	0x3
	sha_se_input	0 → SHA input is treated as fully differential 1 → SHA input is treated as single-ended	[3]	0x0
	sha1_bias	Bias adjustment for SHA1. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	6:4]	0x3

Register	Control Name	Description	Bits	Default = 0x33
5 0x85	sha2_bias	Bias adjustment for SHA2. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	[2:0]	0x3
	sha3_bias	Bias adjustment for SHA3. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	6:4]	0x3

Register	Control Name	Description	Bits	Default = 0x33
6 0x86	sha4_bias	Bias adjustment for SHA4. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	[2:0]	0x3
	sha5_bias	Bias adjustment for SHA5. SHA bias current is nominally 80 μ A – 10 μ A *sha0_bias[2:0]	6:4]	0x3

Register	Control Name	Description	Bits	Default = 0x33
7 0x87	sha6_bias	Bias adjustment for SHA6. SHA bias current is nominally $80\ \mu\text{A} - 10\ \mu\text{A} * \text{sha0_bias}[2:0]$	[2:0]	0x3
	sha7_bias	Bias adjustment for SHA7. SHA bias current is nominally $80\ \mu\text{A} - 10\ \mu\text{A} * \text{sha0_bias}[2:0]$	[6:4]	0x3
	sha_early_clks	Enables clocks that rise one 64 MHz clock cycle early	[7]	0x0

When the SHA is frozen, the ADC will oversample the SHA output by a factor of 8. This may be useful in some applications. Note that each SHA has an independent bias adjustment word (both SHAs corresponding to a different ADC share this word). Independent control may be useful to optimize power consumption since each SHA has different settling requirements.

Pipeline ADC Configuration

Register	Control Name	Description	Bits	Default = 0x0B
8 0x88	adc_bias[2:0]	Bias adjustment for OP Amp diff pair. Nominal for warm is 011, nominal for cold is 101. Cheng-Ju used 011 for all temps. ADC bias current is nominally $80\ \mu\text{A} - 10\ \mu\text{A} * \text{adc_bias}[2:0]$.	[2:0]	0b011
	nonov_ctrl[1:0]	Allows adjustment of non-overlap time of ADC clocks. Use to adjust timing across temperature.	[4:3]	0b01
	edge_select	Chooses edge of master clock to re-time data. 0 → nominal edge 1 → 180 out of phase in case of gross clock skew errors	[5]	0

Register	Control Name	Description	Bits	Default = 0x00
9 0x89	adc_pd[1:0]	0 → normal operation 1 → power down ADC bit [0] corresponds to ADC0 and bit [1] corresponds to ADC1 When powering down an ADC, be sure to power down all corresponding SHAs (e.g. if you set adc_pd[0] to 1, also set sha_pd_0 to 0xFF)	[1:0]	0b00
	adc_disable_gb	0 → normal operation 1 → Gain boosting amplifiers in the ADC MDACs disabled (powered down with Hi-Z output)	[2]	0
	adc_output_format	Chooses format of ADC output codes 0 → two's complement 1 → offset binary	[3]	0
	adc_sync_mode	0 → normal operation 1 → send out known analog pattern for synchronization. The pattern is from SHA0 through SHA7: (VREF, VCMO, -VREF, VCMO, VCMO, VCMO, VCMO, VCMO)	[4]	0
	adc_test_mode	0 → normal operation (ADC input is SHA output) 1 → ADC converts signal applied to ADC_TEST_INPUT_{P/N} pads (SHA is bypassed)	[5]	0
	adc_output_select	Selects what ADC output is sent off-chip. This can be useful in evaluating the performance of the calibration. 00 or 11 → calibrated ADC data 01 → uncalibrated ADC data 10 → raw ADC0 decisions 11 → raw ADC1 decisions	[7:6]	0

Bandgap Reference Configuration

Register	Control Name	Description	Bits	Default = 0xF1
10 0x8A	vrefp_ctrl[7:0]	Settings to generate VREFP. $vrefp = 1.87 + (ioffset - idac) * 40k$ where $idac = 0.2 \mu A * (257 - vrefp_ctrl)$. Default VREFP is 1.95 V.	[7:0]	0xF1

Register	Control Name	Description	Bits	Default = 0x29
11 0x8B	vrefn_ctrl[7:0]	Settings to generate VREFN. $vrefn = 1.87 + (ioffset - idac) * 40k$ where $idac = 0.2 \mu A * (257 - vrefn_ctrl)$. Default VREFN is 0.45 V.	[7:0]	0x29

Register	Control Name	Description	Bits	Default = 0x8D
12 0x8C	vcmo_ctrl[7:0]	Settings to generate VCMO. $vcmo = 1.87 + (ioffset - idac) * 40k$ where $idac = 0.2 \mu A * (257 - vcmo_ctrl)$. Default VCMO is 1.2 V.	[7:0]	0x8D

Register	Control Name	Description	Bits	Default = 0x65
13 0x8D	vcmi_ctrl[7:0]	Settings to generate VCMO. $vcmi = 1.87 + (ioffset - idac) * 40k$ where $idac = 0.2 \mu A * (257 - vcml_ctrl)$. Default VCMO is 0.9 V.	[7:0]	0x65

Register	Control Name	Description	Bits	Default = 0x55
14 0x8E	vrefp_offset[1:0]	Adjusts ioffset for VREFP DAC. $ioffset = 0 \rightarrow 9.5 \mu A$, $1 \rightarrow 6.3 \mu A$, $2 \rightarrow 9.5 \mu A$, $3 \rightarrow 12.6 \mu A$	[1:0]	0b01
	vrefn_offset[1:0]	Adjusts ioffset for VREFN DAC. $ioffset = 0 \rightarrow 9.5 \mu A$, $1 \rightarrow 6.3 \mu A$, $2 \rightarrow 9.5 \mu A$, $3 \rightarrow 12.6 \mu A$	[3:2]	0b01
	vcmo_offset[1:0]	Adjusts ioffset for VCMO DAC. $ioffset = 0 \rightarrow 9.5 \mu A$, $1 \rightarrow 6.3 \mu A$, $2 \rightarrow 9.5 \mu A$, $3 \rightarrow 12.6 \mu A$	[5:4]	0b01
	vcml_offset[1:0]	Adjusts ioffset for VCMO DAC. $ioffset = 0 \rightarrow 9.5 \mu A$, $1 \rightarrow 6.3 \mu A$, $2 \rightarrow 9.5 \mu A$, $3 \rightarrow 12.6 \mu A$	[7:6]	0b01

Register	Control Name	Description	Bits	Default = 0xFF
15 0x8F	ibuff0_ctrl[7:0]	200 μA nominal current source for input buffer. Current is $2 \mu A (257 - i_buff0_ctrl)$.	[7:0]	0xFF

Register	Control Name	Description	Bits	Default = 0xFF
16 0x90	ibuff1_ctrl[7:0]	200 μA nominal current source for input buffer. Current is $2 \mu A (257 - i_buff0_ctrl)$.	[7:0]	0xFF

Register	Control Name	Description	Bits	Default = 0xFF
17 0x91	i_vdac_0_ctrl[7:0]	200 μA nominal current source for VDAC. Current is $2 \mu A (257 - i_vdac_0_ctrl)$.	[7:0]	0xFF

Register	Control Name	Description	Bits	Default = 0xFF
18 0x92	i_vdac_1_ctrl[7:0]	200 μA nominal current source for VDAC. Current is $2 \mu A (257 - i_vdac_1_ctrl)$.	[7:0]	0xFF

Register	Control Name	Description	Bits	Default = 0x04
19 0x93	external_reference	0 → internal references used 1 → external references used	[0]	0
	external_bgr	0 → internal bandgap reference used 1 → external bandgap reference used	[1]	0
	bgr_select	Selects which circuit is internal voltage references. 0 → BJT-based reference 1 → CMOS-based reference	[2]	1

Register	Control Name	Description	Bits	Default = 0x00
20 0x94	ref_monitor[7:0]	Low byte for reference monitor block control. Each control bit is active high. bit[0] → VREF (250 mV) on aux1 bit[1] → VREFN_BJT_EXT on aux1 bit[2] → VREFP_BJT_EXT on aux1 bit[3] → VCM1_BJT_EXT on aux1 bit[4] → VCMO_BJT_EXT on aux1 bit[5] → Isource_vdac0 on aux3 bit[6] → Isource_vdac1 on aux3 bit[7] → Isource_ibuff0 on aux3	[7:0]	0x00

Register	Control Name	Description	Bits	Default = 0x00
21 0x95	ref_monitor[15:8]	High byte for reference monitor block control. Each control bit is active high. Byte is pre-decoded so each bit corresponds to a different control. It is the user's responsibility to ensure only one bit for each aux pad is high at a time. bit[8] → Isource_ibuff1 on aux3 bit[9] → Isink_adc1 (50 μ A) on aux2 bit[10] → Isink_adc0 (50 μ A) on aux2 bit[11] → Isink_sha1 (50 μ A) on aux2 bit[12] → Isink_sha0 (50 μ A) on aux2 bit[13] → Isink_refbuffers_0 (250 μ A) on aux2 bit[14] → Isink_refbuffers_1 (250 μ A) on aux2 bit[15] → VBGR (1.2 V) on aux1	[7:0]	0x00

Register	Control Name	Description	Bits	Default = 0xFF
22 0x96	ref_powerdown[7:0]	Power down low byte for reference blocks 0 → normal operation 1 → power down bit[0] → vref_bjt_ext bit[1] → vrefn_bjt_ext bit[2] → vrefp_bjt_ext bit[3] → vcmi_bjt_ext bit[4] → vcmo_bjt_ext bit[5] → ibias_200u_source_bjt_ibuff1 bit[6] → ibias_200u_source_bjt_ibuff0 bit[7] → ibias_200u_source_bjt_vdac1	[7:0]	0xFF

Register	Control Name	Description	Bits	Default = 0x2F
23 0x97	ref_powerdown[11:8]	Power down high nibble for reference blocks 0 → normal operation 1 → power down bit[8] → ibias_200u_source_bjt_vdac0 bit[9] → all adc + sha bjt currents bit[10] → both refbuffer_bjt currents bit[11] → singled-ended amp bias (internal to BJT-based bias block)	[3:0]	0xF
	ref_bias[2:0]	Bias adjustment for ADC reference buffers. Bias current is nominally 400 μ A – 50 μ A *ref_bias[2:0] Note: ref_bias controls the magnitude of the current used by the ADC reference buffers independent of whether the Bandgap reference block or the CMOS reference block is being used.	[6:4]	0b010

CMOS Reference Configuration

Register	Control Name	Description	Bits	Default = 0xDF
24 0x98	vrefp_ctrl_cmos[7:0]	Setting for VREFP (in case CMOS reference is desired). Output is (ctrl /255) *VDDA2P5 V. Active if bgr_select = 1. Set to 0 if unused.	[7:0]	0xDF

Register	Control Name	Description	Bits	Default = 0x33
25 0x99	vrefn_ctrl_cmos[7:0]	Setting for VREFN (in case CMOS reference is desired). Output is (ctrl /255) * VDDA2P5 V. Active if bgr_select = 1. Set to 0 if unused.	[7:0]	0x33

Register	Control Name	Description	Bits	Default = 0x89
26 0x9A	vcmo_ctrl_cmos[7:0]	Setting for VCMO (in case CMOS reference is desired). Output is (ctrl /255) * VDDA2P5 V. Active if bgr_select = 1. Set to 0 if unused.	[7:0]	0x89

Register	Control Name	Description	Bits	Default = 0x67
27 0x9B	vcmi_ctrl_cmos[7:0]	Setting for VCMO (in case CMOS reference is desired). Output is (ctrl /255) * VDDA2P5 V. Active if bgr_select = 1. Set to 0 if unused.	[7:0]	0x67

Register	Control Name	Description	Bits	Default = 0x15
28 0x9C	iref_sel[1:0]	Choose where chip bias currents generated by the CMOS reference generator come from. 00 → Turn off bias current (bias currents come from BJT-based reference in this case) 01 → CMOS reference with internal R 10 → CMOS reference with external R 11 → plan B reference (external R directly connected to PMOS current source)	[1:0]	0b01
	vt_iref_trim_ctrl[2:0]	Trim for vt-referenced currents. Nominally the vt-reference current will be 70μA - 5 μA * vt_iref_trim_ctrl[2:0], although the constant term will vary a lot with process shifts. Control is active low.	[4:2]	0b101
	vt_kickstart	Forces CMOS reference away from zero-current state. To kickstart the CMOS reference, set vt_kickstart → 1 and then immediately return it to vt_kickstart → 0. Hopefully you will never have to do this.	[5]	0

Register	Control Name	Description	Bits	Default = 0xFF
29 0x9D	ibuff0_cmos[5:0]	Bias current setting for nominal 200 μA source for the input buffers. Current is 512 μA - ibuff0_cmos<5:0>*8 μA .	[5:0]	0xFF

Register	Control Name	Description	Bits	Default = 0xFF
30 0x9E	ibuff1_cmos[5:0]	Bias current setting for nominal 200 μA source for the input buffers. Current is 512 μA - ibuff1_cmos<5:0>*8 μA .	[5:0]	0xFF

Calibration Engine Configuration

Register	Control Name	Description	Bits	Default = 0x0
31 0x9F	calibrate[1:0]	Triggers calibration. Must return to 0 before another calibration can be started. 0 → normal operation 1 → Initiate calibration sequence Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[1:0]	0b00
	load_cal_defaults	Restores calibration weights to default state. 0 → normal operation 1 → zero weights. Default weights are loaded when load_cal_defaults is set back to 0.	[2]	0
	load_config_defaults	Restores ADC configuration to default state. Configuration will only respond to write commands when bit = 0. 0 → normal operation 1 → restore configuration settings to default	[3]	0

Register	Control Name	Description	Bits	Default = 0x7F
32 0xA0	meas_cycles[3:0]	Number of ADC samples taken for each measurement. Number of samples is $2^{(\text{meas_cycles})}$ so the maximum setting is 2^{15} .	[3:0]	0xF
		Bits [6:4] are unused (were originally cal_stages[2:0])	[6:4]	0b111
	test_correction_logic	Puts correction logic into test mode. Test data is stored in configuration registers 33 through 40) 0 → normal operation 1 → set test data through correction logic The test uses a 30-bit decision vector, with one {MSB,LSB} pair per stage. In test word, Bit[14] corresponds to stage 0, bit [0] corresponds to stage 14. Bit[15] is ignored. The test data is included in configuration registers	[7]	0

Register	Control Name	Description	Bits	Default = 0x00
33 0xA1	test_lsb0[7:0]	LSBs (low byte) for correction logic test word, ADC0.	[7:0]	0

Register	Control Name	Description	Bits	Default = 0x00
34 0xA2	test_lsb0[14:8]	LSBs (high byte) for correction logic test word, ADC0.	[6:0]	0

Register	Control Name	Description	Bits	Default = 0x00
35 0xA3	test_msb0[7:0]	MSBs (low byte) for correction logic test word, ADC0.	[7:0]	0

Register	Control Name	Description	Bits	Default = 0x00
36 0xA4	test_msb0[14:8]	MSBs (high byte) for correction logic test word, ADC0.	[6:0]	0

Register	Control Name	Description	Bits	Default = 0x00
37 0xA5	test_lsb1[7:0]	LSBs (low byte) for correction logic test word, ADC1.	[7:0]	0

Register	Control Name	Description	Bits	Default = 0x00
38 0xA6	test_lsb1[14:8]	LSBs (high byte) for correction logic test word, ADC1.	[6:0]	0

Register	Control Name	Description	Bits	Default = 0x00
39 0xA7	test_msb1[7:0]	MSBs (low byte) for correction logic test word, ADC1.	[7:0]	0

Register	Control Name	Description	Bits	Default = 0x00
40 0xA8	test_msb1[14:8]	MSBs (high byte) for correction logic test word, ADC1.	[6:0]	0

Overflow Logic Configuration

Register	Control Name	Description	Bits	Default = 0x01
41 0xA9	enable_overflow_mon	Enables overflow correction logic if set = 1	[0]	1

Ring Oscillator Configuration

Register	Control Name	Description	Bits	Default = 0x00
42 0xAA	enable_ringosc	Enables ring oscillator output to pad if set = 1 Ring oscillator is included as a process monitor.	[0]	0

Register	Control Name	Description	Bits	Default = 0x00
43 0xAB	spare	Unused register	[7:0]	0x00

Calibration Forcing Configuration

The following registers are used to allow calibration to be done off chip. To calibrate off chip, overflow correction must be disabled (by setting *enable_overflow_mon* = 0) while collecting data.

Register	Control Name	Description	Bits	Default = 0x00
44 0xAC	force_enable[1:0]	Put chip into external forcing mode. 0 → normal operation 1 → forcing mode Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[1:0]	0
	stage_select[2:0]	Selects which stage is put into forcing mode. MSB is stage 0, the next most significant stage is stage 1, and so on. Ex: 000 → stage0, 110 → stage6 Setting 111 has no effect.	[4:2]	0
	force_refp[1:0]	Forces conversion of positive differential reference. 0 → no force 1 → force (VREFP-VREFN) at input Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[6:5]	0

Register	Control Name	Description	Bits	Default = 0x00
45 0xAD	force_refn[1:0]	Forces conversion of negative differential reference. 0 → no force 1 → force (VREFN-VREFP) at input Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[1:0]	0
	force_cm[1:0]	Forces conversion of input common-mode reference. 0 → no force 1 → force VCMI at input Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[3:2]	0
	force_msb[1:0]	Force MSB to 1. {MSB,LSB} is stage raw digital output. 0 → force MSB to 0 1 → force MSB to 1 Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[5:4]	0

Register	Control Name	Description	Bits	Default = 0x00
46 0xAE	force_lsb[1:0]	Force LSB to 1. {MSB,LSB} is stage raw digital output. 0 → force LSB to 0 1 → force LSB to 1 Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[1:0]	0
	caldac_ctrl[1:0]	Choose forcing DAC setting 0 → S0, S1 measurement 1 → S2, S3 measurement Bit 0 corresponds to ADC0 and Bit 1 corresponds to ADC1.	[3:2]	0
	clear_regs[1:0]	Clears calibration registers. 0 → ADC0 1 → ADC1	[5:4]	0

Monitor Output Configuration

Register	Control Name	Description	Bits	Default = 0x00
47 0xAF	vmonitor_enable	0 → vmonitor disabled 1 → vmonitor enabled	[0]	0
	imonitor_enable	0 → imonitor disabled 1 → imonitor enabled	[1]	0
	vmonitor_select[2:0]	Selects internal voltage to monitor 000 → VBGR 001 → VCM1 (before ref buffers) 010 → VCM0 (before ref buffers) 011 → VREFP (before ref buffers) 100 → VREFN (before ref buffers) 101 → VBGR 110 → VSSA 111 → VSSA	[4:2]	0
	imonitor_select[2:0]	Selects internal current to monitor 000 → ICMOS_REF 001 → ISHA0 010 → IADC0 011 → ISHA1 100 → IADC1 101 → IBUFF (CMOS) 110 → IREF 111 → IREFBUFFER0 SHA current based on sha_bias0 code. Buffer current based on idiff_buff0 code.	[7:5]	0

Backend Configuration

Register	Control Name	Description	Bits	Default = 0x04
48 0xB0 (page 2 reg 2)	config_lvds_i_ctrl[2:0]	Sets output LVDS current. 000 → 165 μ A 001 → 440 μ A 010 → 715 μ A 011 → 990 μ A 100 → 1260 μ A 101 → 1530 μ A 110 → 1800 μ A 111 → 2070 μ A	[2:0]	0b100

Register	Control Name	Description	Bits	Default = 0x10
49 0xB1 (page 2 reg 1)	config_start_number[4:0]	Number of positive edges of the 64 MHz clock between the 2 MHz clock edge in the backend and the frame start marker.	[4:0]	0x10 <i>NOTE: should be 0x0C</i>
(page 2 reg 4)	config_sha0_pointer	Select which SHA select signal will start a sample period	[7:5]	0b000

Register	Control Name	Description	Bits	Default = 0x00
50 0xB2 (page 2 reg 3)	config_debug_enable	Enables observation of internal digital signals and clocks at output pins.	[0]	0x0
	config_debug_select[3:0]	Selects digital signal to send to debug output. 0000 → 16 MHz ADC clock (internally generated) 0001 → 64 MHz backend clock (loopback) 0010 → 2 MHz sample clock (resampled internally) 0011 → CCW_frontEndSample 0100 → I2C_Ack1 0101 → I2C_Ack1 0110 → I2C_Ack3 0111 → I2C_softReset 1000 → I2C_writeReq 1001 → CCW_write_external_I2C 1010 → CCW_read_external_I2C 1011 → CCW_external_mode_I2C 1100 → Parity (ADC_dataA) 1101 → Parity (ADC_dataB) 1110 → PRBS7 bitstream 1111 → PRBS15 bitstream	[4:1]	0b0000
(not included in page 2 reg 3)	config_test_data_mode	Select ADC data or test patterns 0 → normal operation (ADC data output) 1 → test pattern output	[5]	0

Register	Control Name	Description	Bits	Default = 0xCD
51 0xB3	config_adc0_pattern[7:0]	Low byte of pattern representing ADC0	[7:0]	0xCD

Register	Control Name	Description	Bits	Default = 0xAB
52 0xB4	config_adc0_pattern[15:8]	High byte of pattern representing ADC0	[7:0]	0xAB

Register	Control Name	Description	Bits	Default = 0x34
53 0xB5	config_adc1_pattern[7:0]	Low byte of pattern representing ADC1	[7:0]	0x34

Register	Control Name	Description	Bits	Default = 0x12
54 0xB6	config_adc1_pattern[15:8]	High byte of pattern representing ADC1	[7:0]	0x12

Page 2 Registers (I2C Only)

Register	Control Name	Description	Bits	Default = 0x10
1	config_start_number[4:0]	Number of positive edges of the 64 MHz clock between the 2 MHz clock edge in the backend and the frame start marker.	[4:0]	0x10 <i>NOTE: should be 0x0C</i>

Register	Control Name	Description	Bits	Default = 0x04
2	config_lvds_i_ctrl[2:0]	Sets output LVDS current. 000 → 165 μ A 001 → 440 μ A 010 → 715 μ A 011 → 990 μ A 100 → 1260 μ A 101 → 1530 μ A 110 → 1800 μ A 111 → 2070 μ A	[2:0]	0b100

Register	Control Name	Description	Bits	Default = 0x00
3	config_debug_enable	Enables observation of internal digital signals and clocks at output pins.	[0]	0x0
	config_debug_select[3:0]	Selects digital signal to send to debug output. 0000 → 16 MHz ADC clock (internally generated) 0001 → 64 MHz backend clock (loopback) 0010 → 2 MHz sample clock (resampled internally) 0011 → CCW_frontEndSample 0100 → I2C_Ack1 0101 → I2C_Ack1 0110 → I2C_Ack3 0111 → I2C_softReset 1000 → I2C_writeReq 1001 → CCW_write_external_I2C 1010 → CCW_read_external_I2C 1011 → CCW_external_mode_I2C 1100 → Parity (ADC_dataA) 1101 → Parity (ADC_dataB) 1110 → PRBS7 bitstream 1111 → PRBS15 bitstream	[4:1]	0b0000

Register	Control Name	Description	Bits	Default = 0x00
4	config_sha0_pointer[2:0]	Select which SHA select signal will start a sample period	[2:0]	0b000

Example Control Register Configurations

Default Configuration

At power up and after a hard reset, all control registers and pipeline weights are set to their default values. In this configuration COLDAC_P2 is setup for normal operation using differential inputs and bypassing the input buffers. Reference voltages and currents are provided by the CMOS reference block. Registers that require different settings at different temperatures are set to values appropriate for operation in liquid argon. The pipeline ADCs are configured to operate in the “calibrated” mode, but with default (uncalibrated) weights. The output is in the form of two’s complement integers.

Calibration

The COLDATA_P2 pipeline ADCs can be calibrated by setting *calibrate*[1:0] = 0b11 in register 31 (register 31 = 0x03). This will start both calibration engines. During calibration COLDADC_P2 will ignore its inputs. As soon as calibration is finished, normal operation will resume. If only one bit of *calibrate* is set to 1, only one of the two pipeline ADCs will be calibrated. In order to redo the calibration one must first set *calibrate* = 0.

Output Format

By default, the COLDADC_P2 output is expressed as **16-bit two's complement** (signed) integers. The most negative number is 0x8000 and the most positive number is 0x7FFF. If *adc_output_format* in register 9 is set to 1 (register 9 = 0x08), the output is expressed in offset binary format (this is done simply by complementing the most significant bit). In this case the most negative number is 0x0000 and the most positive number is 0xFFFF.

Channel Order and Digital Frame Marker Alignment

The digital output frame marker should be aligned with the most significant bit in each nibble of information from channel 0 and channel 8. The nibble alignment can be checked by setting *config_test_data_mode* = 1 in register 50 (register 50 = 0x20). The data corresponding to ADC0 should then be 0xABCD and the data corresponding to ADC1 should be 0x1234. The timing of the frame marker can be adjusted using *config_start_number*[4:0]. **NOTE: the default value of 0x10 is wrong; the correct value is 0x0C.** The channel order can be verified by setting *adc_sync_mode* = 1 in register 9 (register 9 = 0x10). This will disable the inputs to the SHAs and instead input VREFP to SHA0 and SHA8, VREFN to SHA2 and SHA10, and VCMO to all other SHAs. The ADC value read out for channel 0 and channel 8 should be close to the maximum possible value, the value read out for channel 2 and channel 10 should be close to the most negative possible value, and all others should be close to 0 (mid-range). The only way to reliably verify that the SHAs are presented to the ADC pipelines in correct order after the input voltages are sampled is to use known waveforms at the normal chip inputs (for instance by pulsing the LArASIC that is being digitized by a COLDADC).